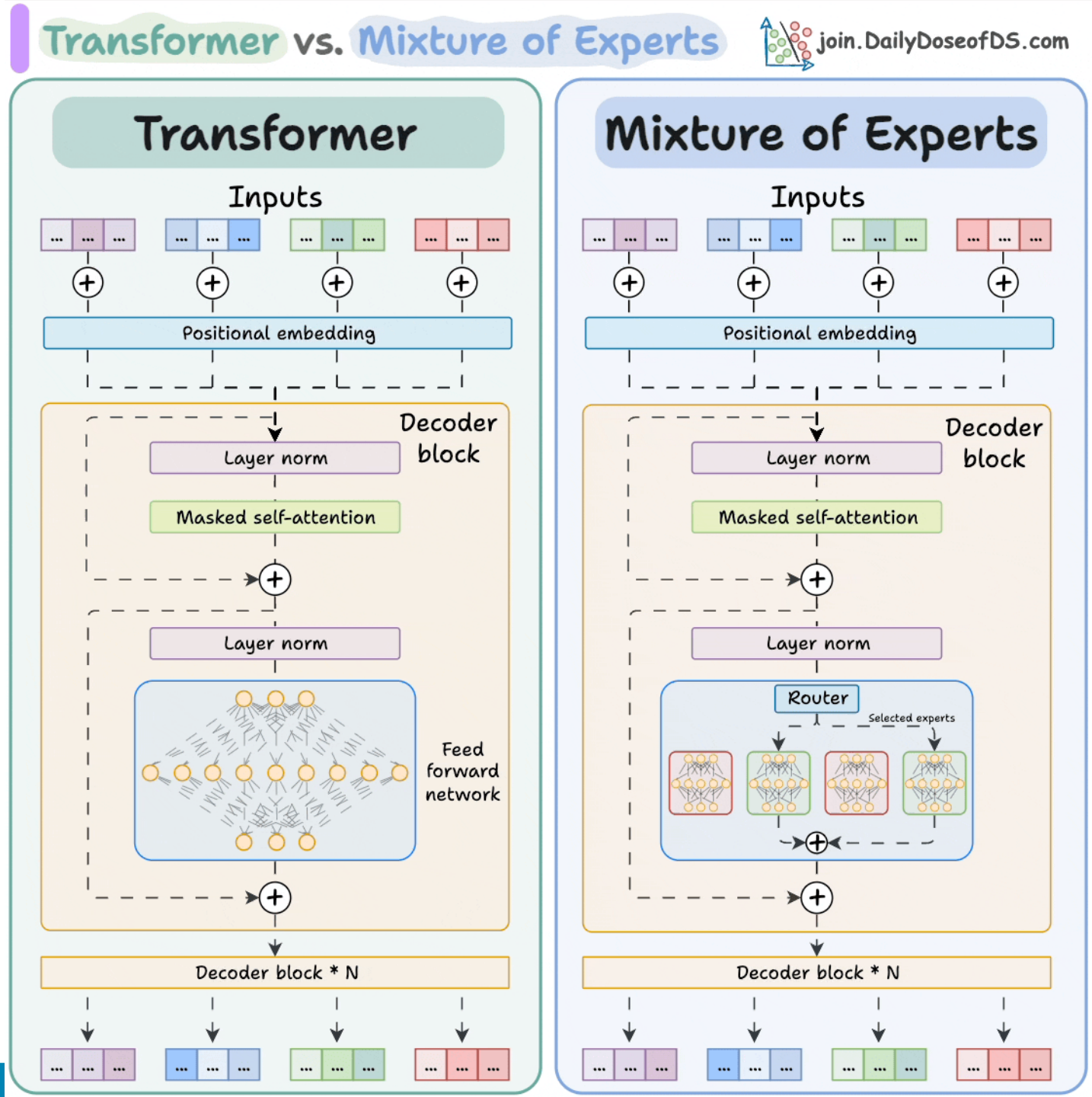
Mixture of Experts

Sparsh Mittal

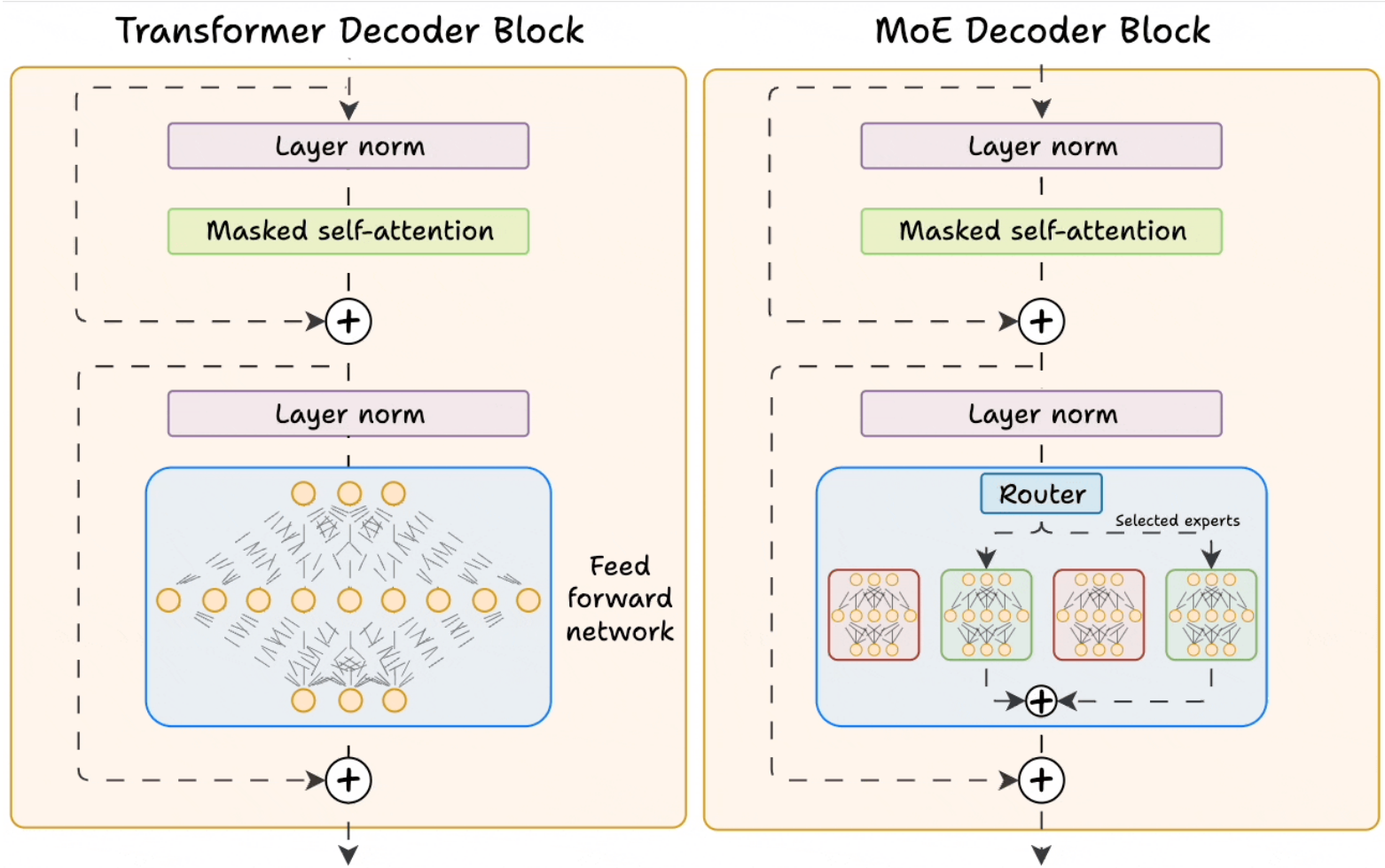


MoE increases model capacity without a proportional increase in computation.

Transformer uses a feed-forward network.

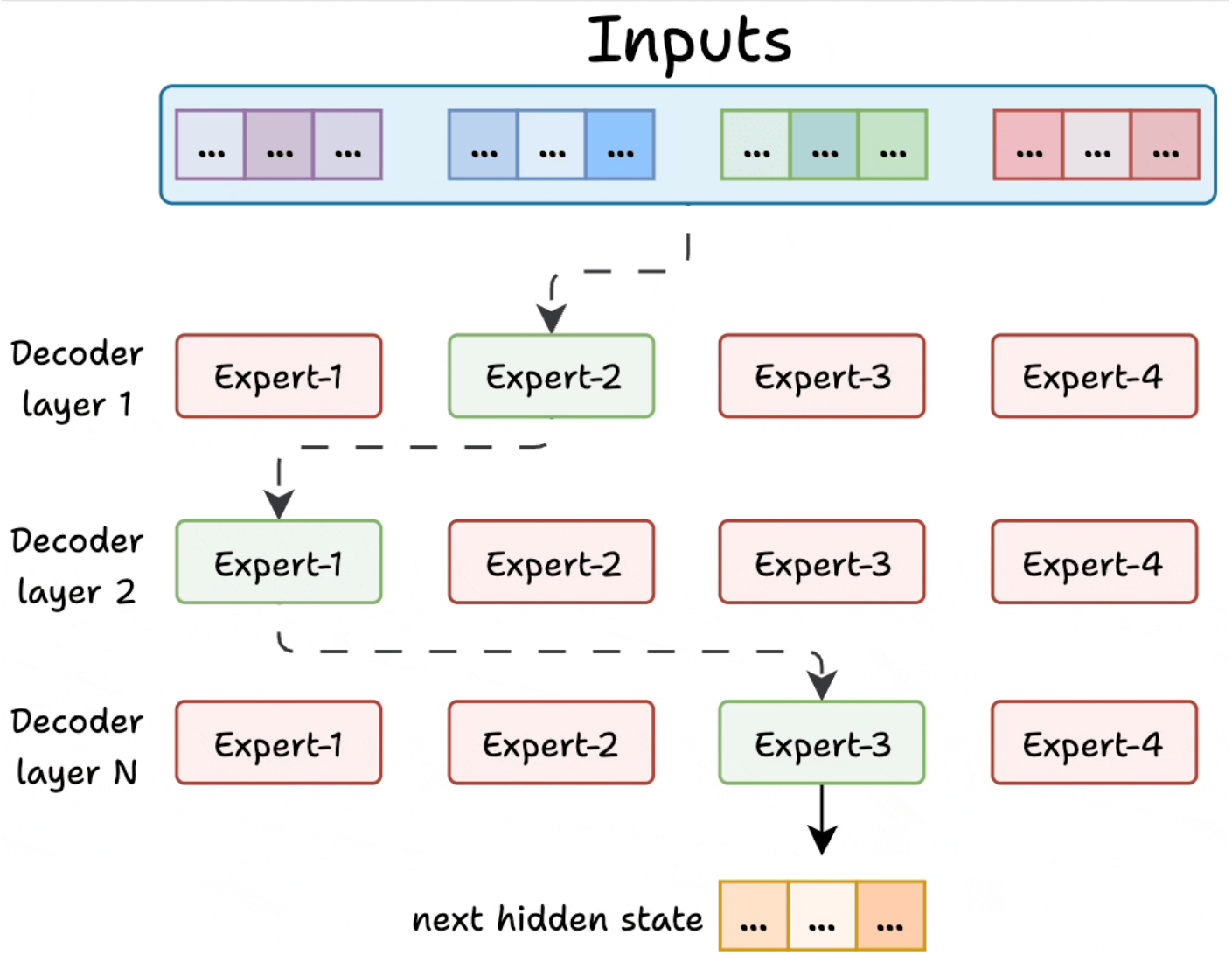
MoE uses experts, which are feed-forward networks but smaller compared to that in Transformer.

During inference, a subset of experts are selected. This makes inference faster in MoE.



Since the network has multiple decoder layers,

- the text passes through different experts across layers.
- the chosen experts also differ between tokens.



Real-world LLMs' examples

- GPT-4's MoE model is believed to house 16 experts, each with around 111 billion parameters each.
- Mixtral employs 8 experts, with only 2 being active at a time.
- DeepSeek utilizes 64 experts, activating 6 at once.
  - This larger, finer-grained expert pool allows for greater specialization and more efficient knowledge acquisition.

HybriMoE: Hybrid CPU-GPU Scheduling and Cache Management for Efficient MoE Inference

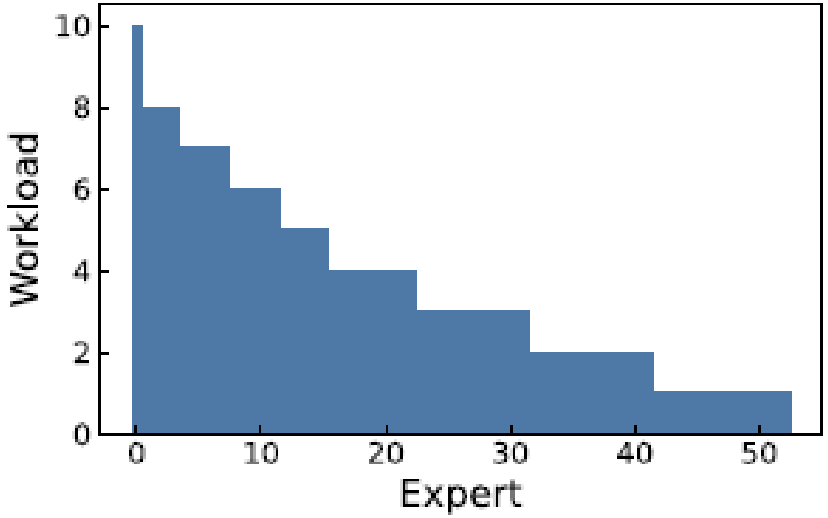
DAC 2025 paper

Challenges of running MoE on CPU-GPU System

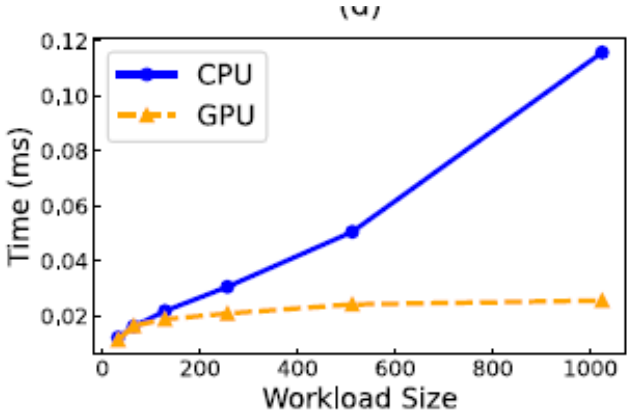
- Large MoE model size introduces substantial memory demands.
- Hybrid CPU-GPU inference has been proposed to leverage CPU computation to reduce expert loading overhead
- However, there are challenges:
- The expert activation patterns of MoE models are highly unstable
  - Fixed mapping strategies become inefficient.
- The hybrid CPU-GPU schedule for MoE is inherently complex due to the diverse expert sizes, structures, uneven workload distribution, etc.

Observations

Different experts have different workloads.

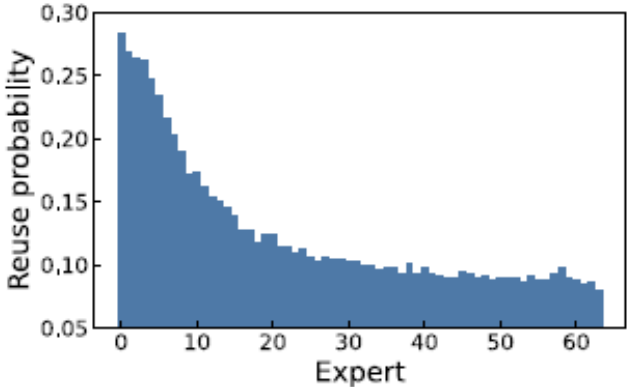


GPU time remains stable with increasing workload, whereas CPU time grows linearly with workload



Idea 1: Score-based caching

Observation: Experts with higher activation scores are more likely to be reused in the next iteration



Idea: Retaining high-score experts in cache can reduce access latency.

Idea 2: Load-based execution

GPU prioritizes computation of cached experts, executing higher-load experts first.

CPU prioritizes computation of uncached experts, focusing on lower-load tasks. Further, CPU can process cached experts when it is idle.

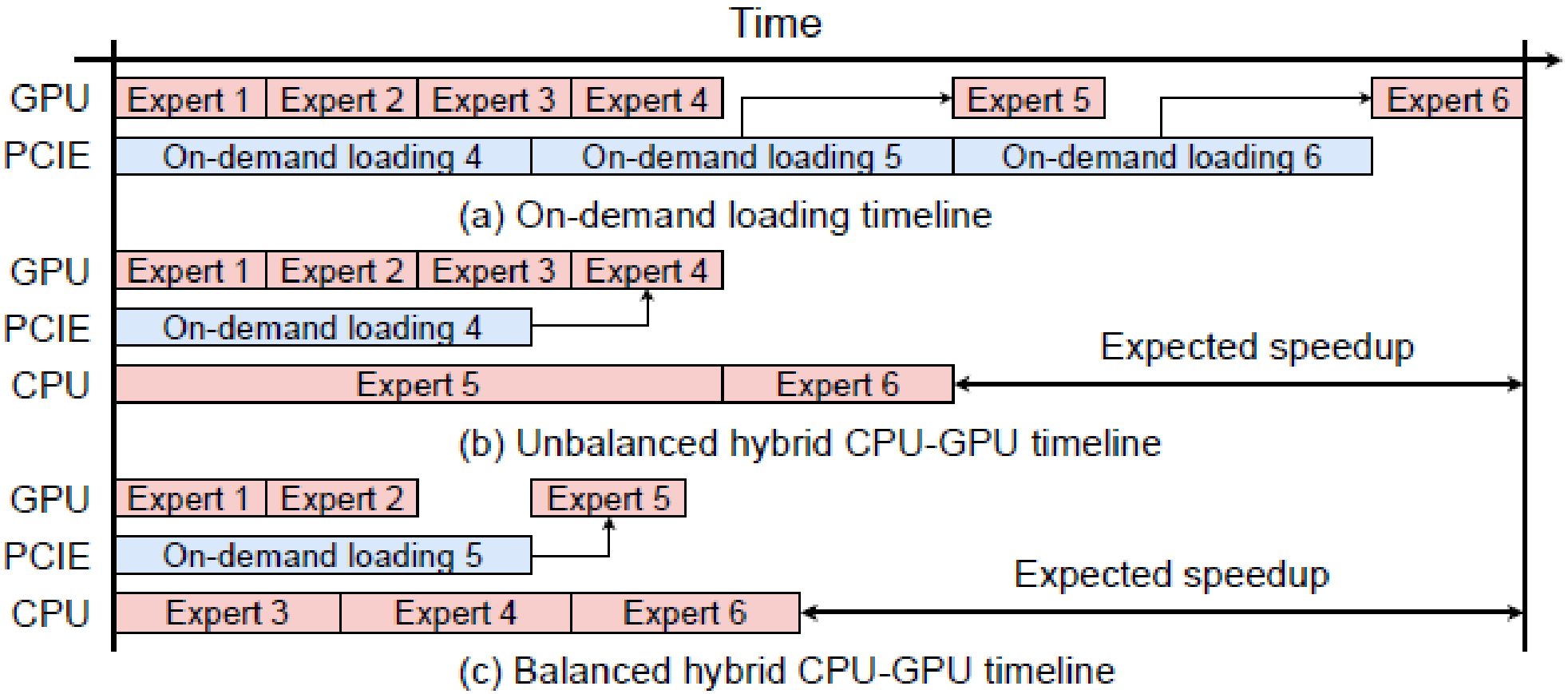
Move high-load uncached experts from CPU to GPU to minimize computation delays.

Idea 3: Impact-based prefetching

Observation: MoE models often exhibit high activation similarity between adjacent layers.

Idea: Predict expert activations for the next three layers.

Then, preload experts likely to be activated in subsequent computations.



Execution timeline of three scenarios. Expert computation time on the GPU remains constant, while CPU execution time increases linearly with workload. The balanced scheduling in (c) achieves improved utilization and reduces overall execution time.